ELEC 221 Tutorial 4.5 – Solutions

DT Complex Exponentials and the DT Fourier Series

Monday, 12 October 2026

Overview

1. Discrete-time complex exponential signals

2. The discrete-time Fourier series

1.1 – Method: periodicity of a sum of harmonics

Each term $\cos(\omega_i n)$ or $e^{j\omega_i n}$ is periodic iff $\omega_i/2\pi$ is rational. Writing $\omega_i/2\pi = p_i/q_i$ in lowest terms (so $\gcd(p_i, q_i) = 1$), the **fundamental period of that term is** $N_i = q_i$: the smallest N with $\omega_i N = 2\pi m$ for integer m is $N = q_i$ (taking $m = p_i$).

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$$N = \mathrm{lcm}(N_1, N_2, \dots)$$

– every term must complete a whole number of cycles simultaneously. If even one term has irrational $\omega_i/2\pi$, that term alone is aperiodic, and no finite N can fix it: the whole sum is **not periodic**.

1.1(a), (b) – Applying the method

(a) $x[n] = 2\cos\left(\frac{\pi}{3}n\right) + \sin\left(\frac{2\pi}{3}n\right)$

- $\omega_1 = \frac{\pi}{3}$: $\dfrac{\omega_1}{2\pi} = \dfrac{1}{6}$ (lowest terms) $\Rightarrow N_1 = 6$
- $\omega_2 = \frac{2\pi}{3}$: $\dfrac{\omega_2}{2\pi} = \dfrac{1}{3}$ (lowest terms) $\Rightarrow N_2 = 3$

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(b) $x[n] = \cos\left(\frac{4\pi}{7}n\right) + e^{j\frac{2\pi}{12}n}$

- $\omega_1 = \frac{4\pi}{7}$: $\dfrac{\omega_1}{2\pi} = \dfrac{2}{7}$ (lowest terms) $\Rightarrow N_1 = 7$
- $\omega_2 = \frac{2\pi}{12} = \frac{\pi}{6}$: $\dfrac{\omega_2}{2\pi} = \dfrac{1}{12}$ (lowest terms) $\Rightarrow N_2 = 12$

$$\boxed{N = \mathrm{lcm}(7, 12) = 84}$$

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- $\omega_1 = \frac{4}{7}$ has **no factor of** π: $\dfrac{\omega_1}{2\pi} = \dfrac{4}{7 \cdot 2\pi} = \dfrac{2}{7\pi}$. Since π is irrational, this ratio is **irrational** – no integer N can ever make $\omega_1 N$ land on a multiple of 2π.
- $\omega_2 = \frac{7\pi}{3}$ alone would be fine ($\omega_2/2\pi = 7/6$, rational, $N_2 = 6$), but it can't rescue the sum: a periodic signal plus a non-periodic one is non-periodic (the irrational-frequency term never re-synchronizes).

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$\boxed{x[n] \text{ is } \textit{not} \text{ periodic}}$

$1.2 - x[n] = e^{-j\pi n/4}$

Fundamental period: $\omega = -\frac{\pi}{4}$ (magnitude $\frac{\pi}{4}$ is what matters for periodicity). $\frac{|\omega|}{2\pi} = \frac{1}{8}$ (lowest terms) $\Rightarrow$

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Positive-frequency equivalent: frequencies in DT only matter modulo 2π, since $e^{j(\omega+2\pi)n} = e^{j\omega n}e^{j2\pi n} = e^{j\omega n}$ for integer n. So add 2π to $-\frac{\pi}{4}$:

$$-\frac{\pi}{4} + 2\pi = \frac{7\pi}{4}$$

$$\boxed{e^{-j\pi n/4} = e^{j(7\pi/4)n}}$$

Check: $\frac{7\pi/4}{2\pi} = \frac{7}{8}$ (lowest terms) $\Rightarrow N = 8$ – same fundamental period, as it must be, since it's the identical sequence just relabelled with a positive-looking frequency.

1.3 – Why harmonics with $k > N/2$ are still needed

It's true that, as an **oscillation shape**, $\cos(\omega n)$ for ω and $2\pi - \omega$ look identical (cosine is even and 2π-periodic) – so frequency "effectively" only ranges up to π.

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- A general real signal needs both members of each conjugate pair, $e^{jk\frac{2\pi}{N}n}$ and $e^{j(N-k)\frac{2\pi}{N}n} = e^{-jk\frac{2\pi}{N}n}$, with *conjugate coefficients* c_k and $c_{N-k} = c_k^*$, so that their imaginary parts cancel and the sum comes out real.
- Keeping only $k = 0, \ldots, N/2$ would throw away exactly the conjugate partners needed to cancel those imaginary parts.

So harmonics $k > N/2$ aren't "faster" oscillations in disguise – they are the negative frequencies (relabelled via $k \leftrightarrow k - N$) that every real signal's spectrum must contain.

2.1 – $\sum_{k=0}^{4} e^{jk2\pi/5} = 0$: rectangular form

With $N = 5$, the five terms sit at angles $0°, 72°, 144°, 216°, 288°$:

k	angle	rectangular form
0	$0°$	$1.0000 + j0.0000$
1	$72°$	$0.3090 + j0.9511$
2	$144°$	$-0.8090 + j0.5878$
3	$216°$	$-0.8090 - j0.5878$
4	$288°$	$0.3090 - j0.9511$

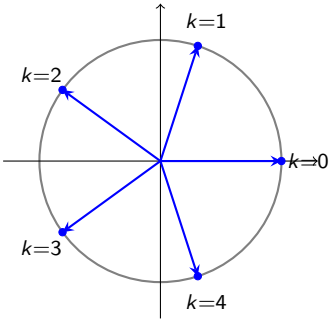
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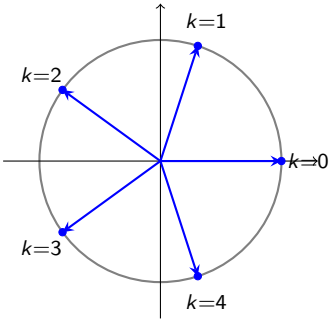
Adding real parts: $1 + 0.3090 - 0.8090 - 0.8090 + 0.3090 = 0$. Adding imaginary parts: the $k = 1, 4$ pair and the $k = 2, 3$ pair are complex conjugates, so their imaginary parts cancel exactly: $0.9511 - 0.9511 + 0.5878 - 0.5878 = 0$.

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The 5 points form a regular pentagon on the unit circle – symmetric about the origin, so the vectors cancel. **General proof (geometric series):** let $r = e^{j2\pi/5} \neq 1$. Then

$$\sum_{k=0}^{4} r^k = \frac{r^5 - 1}{r - 1} = \frac{e^{j2\pi} - 1}{r - 1} = \frac{1 - 1}{r - 1} = 0$$

since $r^5 = e^{j2\pi} = 1$. This works for any $N > 1$: the sum of all N-th roots of unity is always 0.

2.2 – Real $x[n] \Rightarrow c_{-k} = c_k^*$

$$c_k = \frac{1}{N} \sum_{n=\langle N \rangle} x[n]\, e^{-jk\frac{2\pi}{N}n}$$

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Conjugate both sides. Since $x[n]$ is real, $x[n]^* = x[n]$:

$$c_k^* = \frac{1}{N} \sum_{n=\langle N \rangle} x[n]^*\, e^{+jk\frac{2\pi}{N}n} = \frac{1}{N} \sum_{n=\langle N \rangle} x[n]\, e^{-j(-k)\frac{2\pi}{N}n}$$

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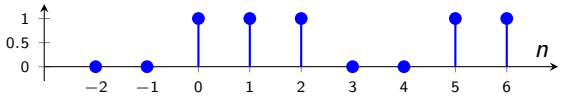
The right-hand side is exactly the analysis formula evaluated at $-k$ instead of k:

$$\boxed{c_k^* = c_{-k}}$$

Simpler than the CT proof in Tutorial 4 – no substitution $t \to -t$ or integral is needed, just conjugating the (finite) sum directly, since $x[n]^* = x[n]$ flips the sign of the exponent for free.

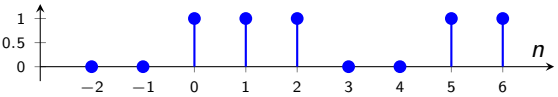
2.3 – DT square wave, $N = 5$: setting up

One period: $x[0] = x[1] = x[2] = 1$, $x[3] = x[4] = 0$.



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Summing $n = 0$ to 4, the zero samples drop out:

$$c_k = \frac{1}{5}\sum_{n=0}^{4} x[n]e^{-jk\frac{2\pi}{5}n} = \frac{1}{5}\left[1 + e^{-jk2\pi/5} + e^{-jk4\pi/5}\right]$$

For $k = 0$: $c_0 = \frac{1}{5}(1 + 1 + 1) = \frac{3}{5}$.

2.3 – Evaluating $c_1, \ldots, c_4$

For $k \neq 0$, using $e^{-j72^\circ} = 0.3090 - j0.9511$ and $e^{-j144^\circ} = -0.8090 - j0.5878$ (and their harmonics):

$$c_1 = \frac{1}{5}\left[1 + (0.3090 - j0.9511) + (-0.8090 - j0.5878)\right] = \frac{1}{5}(0.5 - j1.5389)$$

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$$c_2 = \frac{1}{5}\left[1 + (-0.8090 - j0.5878) + (0.3090 + j0.9511)\right] = \frac{1}{5}(0.5 + j0.3633)$$

$$\boxed{c_2 = 0.1000 + j0.0727}$$

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By conjugate symmetry (2.2), with $c_{-k} = c_{N-k}$ for period N:

$$\boxed{c_3 = c_{-2} = c_2^* = 0.1000 - j0.0727}, \qquad \boxed{c_4 = c_{-1} = c_1^* = 0.1000 + j0.3078}$$

2.3 – Summing over a different window

Now sum over $n = -2, \ldots, 2$ instead. Extending $x[n]$ periodically: $x[-2] = x[3] = 0$ and $x[-1] = x[4] = 0$, while $x[0] = x[1] = x[2] = 1$ – the same five values, just relabelled.

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The coefficients are unchanged

This confirms the rule stated with the DT analysis equation: $x[n]e^{-jk\frac{2\pi}{N}n}$ is itself periodic with period N, so its sum over *any* N consecutive samples – not just $n = 0, \ldots, N-1$ – gives the same total.